Answers & Solutions:

1. **C. essentially remains the same**
2. **D. Diac**
3. **C. has low leakage current**
4. **B. negative-resistance**
5. **A. It should be synchronized with the main supply**
6. **D. Should be removed in order to avoid increased losses and higher junction temperature**
7. **A. SCR**
8. **C. SCS**
9. **C. negative-resistance**
10. **A. 47.5°**

Solution:

$$I_{O(ave)}(2\pi) = \int_{\alpha}^{\pi} I_m \sin\theta d\theta$$
$$I_{O(ave)}(2\pi) = I_m\left[-\cos 180 + \cos\alpha\right]$$
$$I_{O(ave)} = \frac{I_m}{2\pi}\left(1+\cos\alpha\right)$$
$$20 = \frac{75}{2\pi}\left(1+\cos\alpha\right)$$
$$\alpha = 47.5° \rightarrow \text{Ans}$$

11. **C. increases**
12. **C. Induction heating**
13. **C. three**
14. **B. a sawtooth generator**
15. **A. positive**
16. **C. two SCRs in parallel opposition**
17. **A. 19.2 V**

Solution:

$$V_{ave} = \delta V_S = t_{on}\left(f\right)V_S$$
$$= \left(8\times10^{-3}\right)\left(100\times10^{3}\right)\left(24\right)$$
$$V_{ave} = 19.2\text{V} \rightarrow \text{Ans}$$

18. **A.** $\dfrac{R_{B1}}{R_{B1} + R_{B2}}$
19. **D. All of these**
20. **A. break over voltage**
21. **D. ac voltage at supply frequency to ac voltage at load frequency**
22. **A. double-based**
23. **A. four**
24. **C. is decreased**
25. **C. holding current**
26. **B. two pn junction**
27. **B. appropriate gate current**
28. **A. input voltage is continuous and output voltage is discontinuous**
29. **B. bidirectional**
30. **A. 0.75**

Solution:

$$V_{o(ave)} = \frac{V_S}{1-\delta}$$
$$\delta = 1 - \frac{V_S}{V_{o(ave)}}$$
$$= 1 - \frac{12}{48}$$
$$\delta = 0.75 \rightarrow \text{Ans}$$

31. **D. 50 Hz frequency**
32. **D. SUS**
33. **B. is negative peak at time t= storage time t_s**
34. **B. 0.7**
35. **B. unidirectional**
36. **C. is inferior and very significant due to the nature of its application**
37. **D. gate current**
38. **A. two**
39. **B. decreased**
40. **A. Voltage commutation**
41. **C. Both positive and negative**
42. **C. anode current**
43. **A. Connected back to back**
44. **B. thyristor**
45. **C. in inverse-parallel**
46. **A. the advantage of high current at low level gate drive**
47. **D. UJT**
48. **A. 2.24 A**

Solution:

$$V_{rms} = \sqrt{\delta}\left(V_S\right) = \sqrt{t_{on}f}\left(V_S\right)$$
$$= \sqrt{0.2\text{ms}\left(1\text{kHz}\right)}\left(60\right)$$
$$V_{rms} = 26.83\text{ V} \rightarrow \text{Ans}$$
$$I_{rms} = \frac{V_{rms}}{R_L} = \frac{26.83\text{ V}}{12\Omega}$$
$$I_{rms} = 2.24\text{ A} \rightarrow \text{Ans}$$

49. **C. 3-5**
50. **C. positive**

END